

1. Raw Probabilities

Raw Model Output:
 $p_{\theta}(y = 1 \mid \mathbf{X}_i)$

*Contains transient spikes
from sensor noise*

2. Temporal Debouncing

3-Tap Median Filter:
 $p_{\text{smooth}} = \text{Med}(p_{t-1}, p_t, p_{t+1})$

*Eliminates isolated
false-alarm predictions*

3. Threshold Decision

Threshold Rule:
 $\tau = 0.77$

$p > \tau$

DANGER
($\hat{y} = 1$)

$p \leq \tau$

SAFE
($\hat{y} = 0$)